ANALYSIS OF DATA COMMUNICATION IN WEB APPLICATION USING BROWSER DEVELOPER TOOLS

AIM

The aim of this study is to analyze how data is communicated between a web browser and a web server using Browser Developer Tools. It focuses on understanding network requests, responses, HTTP methods, status codes, headers, and data transfer during the use of a web application.

OBJECTIVES

The main objectives of this study are:

1. To understand the concept of data communication in web applications.
2. To study communication between the client (browser) and server.
3. To understand the role of HTTP and HTTPS protocols.
4. To analyze network requests and responses using Browser Developer Tools.
5. To study HTTP methods such as GET, POST, PUT, and DELETE.
6. To understand HTTP response status codes.
7. To identify network errors and performance issues.
8. To understand how Developer Tools help in debugging and improving web applications.

PROBLEM STATEMENT

Web applications depend on continuous communication between the user's browser and a web server. Whenever a user opens a website, searches for information, logs in, submits a form, or performs an online transaction, data is exchanged between the browser and server.

For example, in an e-commerce application, when a user searches for a product, the browser sends a request to the server. The server processes the request and returns the required product information. Similarly, when a user adds a product to a shopping cart, information is sent to the server to update the cart.

Although these operations usually happen very quickly, several requests and responses may take place in the background. If a request fails or takes too long, it can affect the performance and functionality of the application. Common problems include failed requests, incorrect responses, slow server responses, missing resources, and large amounts of transferred data.

It is difficult to identify these problems by simply looking at the webpage. Therefore, a tool is required to monitor and analyze the communication taking place between the browser and server. Browser Developer Tools, particularly the Network panel, provide an effective solution for this purpose.

PROPOSED SOLUTION

The proposed solution is to use Browser Developer Tools to monitor and analyze network communication in a web application. Modern browsers such as Google Chrome, Microsoft Edge, and Mozilla Firefox provide built-in Developer Tools.

The analysis can be performed using the following steps:

Step 1: Open the Web Application

Open the selected web application in a browser. An e-commerce website, login system, or any other web application can be used for analysis.

Step 2: Open Developer Tools

Developer Tools can be opened by pressing F12 or Ctrl + Shift + I. Select the Network tab to view the network activity of the webpage.

Step 3: Perform an Activity

Perform an action such as searching for a product, logging in, submitting a form, or adding an item to a cart. These actions generate network requests.

Step 4: Analyze Network Requests

The Network panel provides information about each request, including:

- Request URL
- HTTP method
- Status code
- Request and response headers
- Request parameters
- Response data
- Transferred data size
- Request duratio**n**

This information helps developers understand how data is exchanged between the browser and server.

Benefits of Browser Developer Tools

- Helps identify failed requests
- Simplifies debugging
- Analyzes HTTP requests and responses
- Detects server/API errors
- Shows transferred data details
- Helps measure page load time
- Improves performance optimization
- Better understanding of client-server communication

CONCLUSION

The analysis of data communication in web applications using Browser Developer Tools provides a practical understanding of how browsers and servers exchange information. By using the Network panel, developers can examine HTTP/HTTPS requests, response status codes, headers, data, and loading times.

This analysis helps identify communication errors, troubleshoot web application problems, and improve application performance. Therefore, Browser Developer Tools are an important and useful resource for **debugging, testing, monitoring, and optimizing modern web applications**.